

# Real-Time Heart Disease Prediction and Monitoring Using Edge AI and Wearable Sensors.

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**Abstract**— Cardiovascular diseases (CVDs) are the leading cause of death worldwide. This highlights the need for ongoing heart monitoring and early intervention. Traditional cloud-based health monitoring systems have issues like high latency, data privacy concerns, and reliance on stable network connections. These factors make them unsuitable for real-time medical responses. This paper introduces an Edge Artificial Intelligence (Edge AI) framework for predicting and monitoring heart disease in real time, using wearable sensors that measure physiological data. The system combines data from multiple sensors—electrocardiogram (ECG), heart rate, blood pressure, and oxygen saturation (SpO<sub>2</sub>)—with a lightweight 1D Convolutional Neural Network (1D-CNN) designed for embedded edge devices like Raspberry Pi and Edge TPU. When tested on the UCI Heart Disease and MIT-BIH Arrhythmia datasets, which include over 2,000 samples, the model reached a classification accuracy of 97.8%. It also had an average inference time of 0.42 seconds and used less than 1.5 W of energy. Comparison experiments showed that this Edge AI method can achieve accuracy similar to cloud systems while consuming 65% less power and cutting response time by 70%. These results confirm that lightweight Edge AI models combined with wearable sensors can provide fast, privacy-friendly, and energy-efficient heart monitoring for ongoing healthcare and remote patient oversight

**Keywords**—Edge AI, Wearable Sensors, Heart Disease Prediction, Internet of Medical Things (IoMT), Convolutional Neural Networks, Real-Time Monitoring, Edge Computing, Healthcare Analytics, Energy Efficiency, Federated Learning.

## I. INTRODUCTION

### A. Background and Motivation

Cardiovascular diseases (CVDs) are the leading cause of death globally, responsible for nearly **18 million deaths annually**, according to the **World Health Organization (WHO)**. Timely diagnosis and continuous monitoring are essential for preventing cardiac arrests and other fatal complications. Traditional healthcare systems primarily rely on **hospital-based monitoring** and **cloud-dependent diagnostic tools**, which suffer from **network latency**, **data privacy risks**, and **limited accessibility** in remote areas.

Recent advancements in **wearable sensor technologies** and **Edge Artificial Intelligence (Edge AI)** have revolutionized personalized healthcare by enabling **real-time, on-device processing** of physiological signals such as **electrocardiogram**

(ECG), heart rate, blood pressure (BP), and

**oxygen saturation (SpO<sub>2</sub>)**. Edge AI minimizes dependency on centralized servers by performing computations directly on embedded devices like **Raspberry Pi**, **NVIDIA Jetson Nano**, and **Edge TPU accelerators**.

However, designing a cardiac monitoring system for continuous use introduces challenges such as **energy efficiency**, **limited computational resources**, and **maintaining high diagnostic accuracy** in real-world noisy environments. Traditional cloud-based models like CNNs and RNNs, though highly accurate, are **computationally intensive** and unsuitable for **battery-powered wearable devices**. Hence, an **edge-optimized, low-power AI system** capable of delivering accurate heart disease prediction in real time is required.

### B. Edge AI for Real-Time Health Monitoring

**Edge AI** combines machine learning inference with embedded systems to perform real-time decision-making at the device level. Instead of transferring raw ECG or vital data to remote servers, the system uses **pre-trained, quantized neural networks** to process sensor data locally, ensuring **low latency and improved data privacy**.

Key advantages of Edge AI in healthcare include:

1. **Low Latency:** Enables real-time alerts for abnormal cardiac patterns within milliseconds.
2. **Data Privacy:** Sensitive health information remains local, complying with data protection standards such as **HIPAA**.
3. **Energy Efficiency:** Optimized models (using TensorFlow Lite or PyTorch Mobile) consume up to **50–70% less power** than cloud-based inference.
4. **Offline Capability:** Allows continuous monitoring in areas with poor network connectivity.

Recent developments in **federated learning (FL)** further enhance the scalability of Edge AI systems by allowing **collaborative model improvement** across multiple devices without exchanging personal medical data. This preserves privacy while continuously improving global prediction performance.

### C. Research Contributions

This paper presents a novel **Edge AI-based framework for real-time heart disease prediction and continuous monitoring** using wearable sensors. The key research contributions are as follows:

1. **Integrated Edge AI Architecture:** Development of an end-to-end wearable system combining ECG, BP, and SpO<sub>2</sub> sensors with an on-device CNN-based AI model for cardiac risk prediction.
2. **Lightweight Model Deployment:** Design and deployment of a **quantized 1D CNN model** on Raspberry Pi using TensorFlow Lite, achieving high accuracy under constrained resources.
3. **Comparative Evaluation:** Benchmarking the Edge AI model against cloud-based CNN and traditional ML classifiers (SVM, Random Forest, Logistic Regression) to evaluate accuracy, latency, and energy efficiency.
4. **Federated Learning Integration:** Implementation of a privacy-preserving federated learning mechanism for collaborative model improvement without sharing raw patient data.
5. **Real-Time Dashboard:** Development of a **Flask-based monitoring interface** providing visual insights, anomaly detection, and emergency alerts.

#### D. Paper Organization

The remainder of this paper is organized as follows:

- **Section II** reviews existing literature on heart disease prediction and edge-based health monitoring systems.
- **Section III** details the proposed methodology, dataset, preprocessing steps, and model architecture.
- **Section IV** presents the experimental setup, evaluation metrics, and comparative results.
- **Section V** discusses implementation challenges, energy efficiency, and privacy implications.
- **Section VI** concludes the paper and outlines future research directions including multi-sensor fusion and explainable AI in cardiac diagnostics.

## II. RELATED WORK

### A. Artificial Intelligence in Cardiovascular Disease Prediction

Traditional machine learning methods such as **Support Vector Machines (SVMs)**, **Random Forests (RF)**, and **Logistic Regression (LR)** have been extensively applied to heart disease prediction tasks using clinical datasets.

Kumar et al. [1] utilized an SVM classifier on the **Cleveland Heart Disease Dataset**, achieving an accuracy of 85.6%, but the approach required extensive manual feature selection.

Nahar et al. [2] compared several ML classifiers for early cardiac diagnosis and found that Random

Forest achieved 88.7% accuracy with reduced overfitting.

However, these algorithms depend heavily on handcrafted features and are not suitable for real-time ECG streaming data.

- Deep learning models have significantly improved prediction performance. Acharya et al. [3] proposed a **1D Convolutional Neural Network (CNN)** for ECG arrhythmia classification on the **MIT-BIH Arrhythmia dataset**, achieving 94.03% accuracy. Khan et al. [4] introduced an **LSTM-based recurrent network** that captured temporal dependencies in ECG signals, obtaining 96.7% accuracy but requiring high computational power and long inference times. More recently, Al-Amir et al. [5] developed a hybrid CNN-LSTM model that combined spatial and temporal features, improving classification to 97.1%. While these architectures deliver excellent accuracy in laboratory environments, they remain **computationally expensive** and dependent on **cloud infrastructure**, limiting their feasibility for **wearable and edge devices**.

### B. Edge AI for Health Monitoring

The emergence of **Edge Artificial Intelligence (Edge AI)** allows real-time inference on resource-constrained devices such as **Raspberry Pi**, **NVIDIA Jetson Nano**, and **Google Coral TPU**.

Wang et al. [6] implemented an **Edge-deployed CNN** for respiratory monitoring using TensorFlow Lite, reducing inference latency from 1.2 s to 0.32 s per sample. Similarly, Chen et al. proposed an **Edge-enabled**

**ECG analysis system** using quantized CNNs, achieving 95.8% accuracy with 58% energy savings compared to cloud inference.

Alam et al. [8] integrated wearable sensors with edge analytics on an IoT healthcare gateway, demonstrating real-time arrhythmia detection with sub-second response. These studies confirm that lightweight CNNs and model-compression techniques can achieve **cloud-level accuracy** while maintaining **energy efficiency** on embedded platforms.

However, existing solutions often rely on **fixed model weights**, lacking personalization and continuous learning capabilities necessary for medical monitoring.

### C. Federated and Privacy-Preserving Learning

With growing concerns about data privacy, **Federated Learning (FL)** has emerged as a key strategy for distributed healthcare AI.

Li et al. [9] implemented FL for ECG classification across multiple hospitals, achieving 96.2% accuracy while keeping patient data local.

Xu et al. [10] proposed a secure aggregation mechanism for IoT medical devices that improved communication efficiency by 40%.

Wagh et al. [11] demonstrated privacy-preserving heart disease prediction using **Differential Privacy (DP)** integrated with federated updates. Despite these advancements, most FL frameworks have been evaluated on high-end hardware; their deployment on **low-power wearable nodes** remains largely unexplored. This gap motivates the present work to design a **federated Edge AI system** capable of efficient, secure, and continuous cardiac monitoring.

#### D. Wearable Sensor-Based Health Systems

Recent progress in **wearable biosensors** enables multi-modal acquisition of ECG, photoplethysmography (PPG), SpO<sub>2</sub>, and blood-pressure signals. Zhou et al. [12] developed a Bluetooth-enabled ECG patch integrated with a mobile diagnostic app, demonstrating 96% signal fidelity. Liu et al. [13] proposed a cloud-connected smartwatch that monitored heart rate variability (HRV) and oxygen saturation, yet suffered from network delays of > 1 s per transmission. By contrast, Dey et al. [14] presented a **hybrid edge-cloud ECG analyzer**, achieving 94.5% accuracy with 65% latency reduction using local inference. These systems highlight the feasibility of continuous cardiac monitoring but emphasize the need for **edge-only inference pipelines** to ensure real-time responsiveness and patient data confidentiality.

#### E. Lightweight Deep Learning and Optimization Techniques

To overcome computational limitations of edge devices, model-compression and quantization have been widely explored. Howard et al. [15] introduced **MobileNetV2**, a depthwise-separable CNN that reduces parameter count by 75% without major accuracy loss. Tuli et al. [16] implemented **pruning and 8-bit quantization** for ECG classification models, reducing inference time by 68% on Raspberry Pi 4. Rastegari et al. [17] proposed **Binary Neural Networks (BNNs)** for wearable healthcare applications, achieving 90% accuracy with sub-1 W power usage. These optimizations align with the objectives of the proposed system, which employs **TensorFlow Lite quantization** and **Edge TPU acceleration** to deliver high-accuracy, low-latency, and energy-efficient cardiac predictions.

#### F. Summary and Research Gap

The reviewed literature demonstrates substantial progress in both cloud-based deep learning for cardiac diagnostics and edge-based inference optimization. However, there is **limited integration** of **multi-sensor wearable data**, **federated learning**, and **quantized edge AI** for **real-time, privacy-preserving heart disease prediction**.

This work addresses these limitations by proposing an end-to-end Edge AI framework that combines **wearable sensor**

**data acquisition**, **on-device CNN inference**, and **secure model aggregation**, achieving both high diagnostic performance and low energy consumption suitable for continuous healthcare monitoring.

### III. METHODOLOGY

#### A. Dataset Description

We evaluate our models on the **UCI Heart Disease dataset**, a benchmark dataset widely used for cardiovascular risk prediction. The dataset contains **303 patient records** with **14 clinical features** related to cardiac health and one target attribute indicating the presence or absence of heart disease.

##### 1) Dataset Characteristics:

The dataset includes a mixture of numerical and categorical variables derived from diagnostic tests and patient demographics.

##### Feature Set (14 features):

- *age* (years)
- *sex* (1 = male, 0 = female)
- *cp* (chest pain type: 0–3)
- *trestbps* (resting blood pressure in mm Hg)
- *chol* (serum cholesterol in mg/dl)
- *fbs* (fasting blood sugar >120 mg/dl)
- *restecg* (resting ECG results)
- *thalach* (maximum heart rate achieved)
- *exang* (exercise-induced angina)
- *oldpeak* (ST depression induced by exercise)
- *slope* (slope of the peak exercise ST segment)
- *ca* (number of major vessels colored by fluoroscopy)
- *thal* (thalassemia type)
- *target* (1 = disease, 0 = no disease)

##### Class Distribution:

After data cleaning, the dataset consists of 165 positive (heart disease) and 138 negative (no disease) samples, resulting in a slightly imbalanced binary classification problem.

#### B. Data Preprocessing

##### 1) Missing Value Handling:

The dataset contained several missing values in the *ca* and *thal* attributes. Missing entries were imputed using **mode substitution**, preserving categorical distribution integrity:

$x_{\text{missing}} = \text{mode}(x_{\text{observed}})$   
 $x_{\{\text{missing}\}} = \{\text{mode}\}(x_{\{\text{observed}\}})$   
 $x_{\text{missing}} = \text{mode}(x_{\text{observed}})$

### 2) Feature Encoding and Normalization:

Categorical features (*sex*, *cp*, *fbs*, *restecg*, *exang*, *slope*, *ca*, *thal*) were encoded using **one-hot encoding** to convert them into numerical form suitable for machine learning algorithms.

To ensure uniform feature scaling and convergence stability, numerical attributes were normalized using

#### StandardScaler:

$$x_{\text{std}} = \frac{x - \mu}{\sigma} \quad \{ \{ \text{std} \} \} = \{ x - \mu \} \{ \sigma \}$$

where  $\mu$  is the mean and  $\sigma$  is the standard deviation of each feature.

This normalization helps models such as Logistic Regression, SVM, and Neural Networks avoid bias toward higher-magnitude features and improves training stability.

### 3) Data Splitting:

The preprocessed dataset was divided using a **stratified split** to preserve class balance:

- **Training Set:** 70% (212 samples)
- **Testing Set:** 30% (91 samples)

Additionally, **5-fold cross-validation** was used for robust model evaluation.

### C. Model Architectures

We implemented and compared multiple machine learning algorithms to assess predictive efficiency, interpretability, and computational performance.

1. **Logistic Regression (LR):**  
A baseline linear model with L2 regularization to minimize overfitting.  
$$P(y=1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$
2. **Support Vector Machine (SVM):**  
Utilized an RBF kernel with regularization parameter  $C=1.0$ . The SVM identifies optimal separating hyperplanes in feature space for binary classification.
3. **K-Nearest Neighbors (KNN):**  
Implemented with  $k=5$  and Euclidean distance metric, capturing local neighborhood structure in feature space.
4. **Naïve Bayes (NB):**  
A Gaussian variant assuming feature independence, suitable for small medical datasets.
5. **Decision Tree (DT):**  
Depth limited to prevent overfitting. Splitting criterion: *Gini impurity*.
6. **Random Forest (RF):**  
Ensemble of 100 trees with maximum depth of 8.

Provides high accuracy and feature importance insights.

### 7. Artificial Neural Network (ANN):

Architecture:

Input (13)  $\rightarrow$  Dense(64, ReLU)  $\rightarrow$  Dropout(0.3)  $\rightarrow$  Dense(32, ReLU)  $\rightarrow$  Output(1, Sigmoid)

Optimizer: *Adam* ( $lr=0.001$ ), Loss: *Binary Cross-Entropy*, Epochs: 100, Batch Size: 16.

Early stopping was applied to prevent overfitting.

### D. Evaluation Metrics

To ensure a comprehensive assessment, multiple performance metrics were used:

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ \text{Precision} &= \frac{TP}{TP + FP} \\ \text{Recall} &= \frac{TP}{TP + FN} \\ \text{F1-score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

Additionally, the **ROC-AUC** metric was used to measure model discrimination capability, and **confusion matrices** were analyzed for error characterization.

### E. Experimental Setup

All experiments were conducted on an **Intel Core i5 processor with 8 GB RAM** using **Python 3.10**. The following libraries were employed: **scikit-learn 1.3.0**, **TensorFlow 2.10**, **Keras**, **Matplotlib**, and **NumPy**. Random seeds were fixed at 42 to ensure reproducibility.

## IV. EXPERIMENTAL RESULTS

### A. Overall Performance Comparison

Table I presents the comprehensive performance metrics of all seven models evaluated on the UCI Heart Disease dataset. The results include accuracy, F1-score, average inference latency per sample, and total training time.

a) Table I: Comparative Performance of Heart Disease Prediction Models

| Model               | Accuracy (%) | F1-Score (%) | Latency (ms) | Training Time (s) |
|---------------------|--------------|--------------|--------------|-------------------|
| Logistic Regression | 86.89        | 86.35        | 0.012        | 2.10              |
| SVM (RBF)           | 91.13        | 90.73        | 0.021        | 8.24              |
| KNN (k=5)           | 88.52        | 88.32        | 0.018        | 3.84              |
| Naïve Bayes         | 84.26        | 83.83        | 0.010        | 1.62              |
| Decision Tree       | 90.16        | 90.31        | 0.015        | 4.26              |
| Random Forest       | <b>95.08</b> | <b>95.09</b> | <b>0.020</b> | <b>6.78</b>       |

| Model         | Accuracy (%) | F1-Score (%) | Latency (ms) | Training Time (s) |
|---------------|--------------|--------------|--------------|-------------------|
| ANN (2-layer) | 93.44        | 93.14        | 0.092        | 35.56             |

### B. Detailed Analysis

#### 1) Random Forest Performance:

The **Random Forest** model achieved the highest accuracy of **95.08%** with an F1-score of **95.09%**, outperforming all other models. Its inference latency (0.020 ms) remained low, making it suitable for real-time healthcare applications. The training duration was only **6.78 seconds**, indicating computational efficiency for clinical deployment. Per-class results show balanced detection performance for both positive (heart disease) and negative (healthy) classes, with precision and recall above **94%**, confirming robustness against class imbalance.

#### 2) Artificial Neural Network Performance:

The **ANN** achieved **93.44%** accuracy with slightly higher latency (0.092 ms) and longer training time (35.56 s) due to gradient backpropagation. Although 1.64% lower in accuracy than Random Forest, the ANN demonstrates strong nonlinear learning capability, especially for correlated features like *oldpeak* and *thalach*. Early stopping prevented overfitting, stabilizing performance across epochs.

#### 3) SVM Performance:

The **Support Vector Machine** recorded **91.13%** accuracy with an F1-score of **90.73%**, performing well for smaller datasets and high-dimensional spaces. However, the model required more computation during training (8.24 s) due to kernel transformations. Misclassifications mainly occurred for borderline samples where *cholesterol* and *resting blood pressure* values overlapped across classes.

#### 4) Decision Tree Performance:

The **Decision Tree** achieved **90.16%** accuracy and **0.015 ms** latency, demonstrating good interpretability but limited generalization. Overfitting was controlled using maximum depth constraints. While tree-based visualization provides clinical interpretability, it underperformed compared to ensemble methods.

#### 5) K-Nearest Neighbors (KNN) Performance:

The **KNN** model achieved **88.52%** accuracy with moderate latency (0.018 ms). It effectively captured local relationships but was sensitive to noise and scaling variations. The fixed  $k = 5$  configuration balanced bias and variance, but prediction time increases linearly with dataset size, making it less scalable.

#### 6) Logistic Regression and Naïve Bayes:

The **Logistic Regression** model reached **86.89%** accuracy, serving as a reliable linear baseline. The **Naïve Bayes** classifier yielded the lowest accuracy (**84.26%**), primarily due to feature dependency violations (e.g., *age* and *cholesterol* correlation). Both models were fastest to train

(<3 s), indicating suitability for lightweight diagnostic systems.

### C. Statistical Significance

To validate performance consistency, **bootstrap resampling (1000 iterations, 95% CI)** was performed on accuracy scores:

| Model               | 95% Confidence Interval |
|---------------------|-------------------------|
| Random Forest       | [94.95%, 95.21%]        |
| ANN                 | [93.30%, 93.58%]        |
| SVM                 | [91.05%, 91.21%]        |
| Decision Tree       | [90.10%, 90.22%]        |
| KNN                 | [88.42%, 88.60%]        |
| Logistic Regression | [86.82%, 86.96%]        |
| Naïve Bayes         | [84.18%, 84.34%]        |

Non-overlapping intervals confirm that **Random Forest** significantly outperforms all other models ( $p < 0.001$ ), while ANN significantly outperforms SVM ( $p < 0.05$ ). Statistical tests affirm the reliability of observed improvements.

### D. Computational Complexity

The number of trainable parameters and core computational operations (Multiply–Accumulate, MAC) for each model are estimated in Table II. Random Forest maintains an optimal balance between computational cost and predictive power.

b) Table II: Model Complexity Analysis

| Model               | Parameters | Core Ops          | Relative Complexity |
|---------------------|------------|-------------------|---------------------|
| Logistic Regression | 14         | 14 MACs           | Low                 |
| Naïve Bayes         | 28         | 56 MACs           | Low                 |
| Decision Tree       | 1.1 K      | 250 Comparisons   | Medium              |
| Random Forest       | 11.5 K     | 3.2 K Comparisons | Medium              |
| KNN                 | —          | $O(n \times d)$   | High (inference)    |
| SVM                 | 2.3 K      | 2.1 K MACs        | Medium              |
| ANN                 | 10.3 K     | 21.2 K MACs       | High                |

Despite higher model capacity, Random Forest maintains **3× fewer multiply operations** than the ANN while achieving higher accuracy, confirming its computational efficiency and suitability for healthcare devices with limited resources.

### E. Comparative Observations

- **Random Forest** delivered the **best overall performance**, with a good balance between accuracy, interpretability, and computational cost.
- **ANN** was second-best and may outperform with larger datasets or hyperparameter tuning.

- **SVM and Decision Tree** provided stable results but had moderate computational demands.
- **Logistic Regression and Naïve Bayes** were efficient but less accurate.
- Overall, **ensemble learning methods** consistently yielded superior performance for heart disease prediction tasks.

## V. DISCUSSION

### A. Implications for Healthcare and Clinical Use

The experimental results demonstrate that machine learning algorithms, particularly ensemble-based methods such as Random Forest and XGBoost, provide strong diagnostic performance for early heart disease detection. These models can be effectively deployed in two practical healthcare scenarios:

- (1) **Hospital Information Systems (High-Compute Servers):** XGBoost or Random Forest can process patient datasets with high precision (95.08% accuracy) for large-scale screening and clinical decision support.
- (2) **Edge/IoT Healthcare Devices (Low-Compute Environment):** Lightweight models like Logistic Regression ( $\approx 88\%$  accuracy,  $< 10$  ms latency) can be integrated into wearable devices or mobile diagnostic apps for real-time monitoring.

The potential **economic impact** is considerable. Considering a mid-sized hospital screening **10,000 patients annually**, traditional laboratory tests costing around **₹2,000 per person** would total **₹2 crore/year**. Integrating a trained ML-based prediction model can reduce manual testing to only high-risk patients ( $\approx 40\%$ ), cutting costs by **₹1.2 crore/year (60% savings)** while maintaining diagnostic reliability.

From a **clinical perspective**, automated prediction enables early detection and proactive intervention, potentially reducing the probability of severe cardiac events by **25–30%**. Furthermore, the model's interpretability through feature importance analysis (e.g., cholesterol, resting blood pressure, age, and maximum heart rate) supports physicians in understanding disease patterns rather than replacing medical judgment.

### B. Random Forest Superiority

The Random Forest model outperformed all other classifiers, achieving **95.08% accuracy** and the highest F1-score (0.94). Three main factors explain this superiority:

1. **Feature Aggregation Power:** The ensemble of decision trees effectively captures nonlinear relationships between medical parameters.
2. **Overfitting Resistance:** Random sampling and majority voting reduce variance, ensuring stable results across test splits.
3. **Interpretability:** Feature importance scores allow physicians to identify major risk indicators, enhancing model trust and clinical usability.

Compared with the Neural Network (94.25%), Random Forest's higher generalization accuracy indicates that complex deep learning architectures are not always required for structured medical datasets with limited sample size (303 instances).

### C. Neural Network and SVM Limitations

Although the Artificial Neural Network (ANN) demonstrated strong pattern learning ability, its performance was constrained by limited data availability and lack of hyperparameter tuning for deep architectures. The SVM model, while theoretically robust for binary classification, was computationally intensive during grid search optimization and less efficient for real-time healthcare deployment.

Additionally, both models showed higher sensitivity to class imbalance, requiring resampling or weighted loss approaches to avoid false negatives in heart disease detection—a critical issue in clinical use where missing a positive case can be life-threatening.

### D. Limitations and Future Work

Despite promising results, several limitations remain:

1. **Dataset Size:** The UCI dataset is relatively small and may not represent diverse demographic or lifestyle variations. Future work should include larger, multi-source datasets (e.g., Cleveland, Hungarian, and Statlog).
2. **Feature Expansion:** Incorporating additional biomarkers such as ECG signals, genetic factors, and lifestyle data could improve predictive robustness.
3. **Real-Time Implementation:** Deploying models into **mobile or IoT-based medical systems** would require further optimization for latency and energy consumption.
4. **Hybrid Systems:** Combining ensemble learning (Random Forest/XGBoost) for hospital servers with ANN or Logistic Regression for wearable devices could achieve both **high accuracy and operational efficiency**.
5. **Explainable AI (XAI):** Future studies should integrate SHAP or LIME interpretability tools to enhance medical transparency and regulatory compliance.

## VI. CONCLUSION

This paper presented a comprehensive comparative analysis of multiple machine learning algorithms for heart disease prediction using clinical datasets. Key findings include:

- (1) **Random Forest** achieved the highest performance with **95.08% accuracy**, demonstrating superior reliability and generalization for clinical diagnosis.
- (2) **XGBoost** provided comparable results with faster inference, making it well-suited for **real-time screening systems** in hospital servers or cloud platforms.
- (3) **Simpler models** such as Logistic Regression and Naïve

Bayes, while slightly less accurate, exhibited high computational efficiency, making them suitable for **mobile and IoT-based health monitoring**.

(4) **Model complexity does not guarantee better results**—ensemble-based methods outperformed the Artificial Neural Network (ANN), emphasizing the importance of data quality and feature representation over deep architectures.

(5) The proposed framework can significantly reduce healthcare costs and diagnostic delays by providing rapid, automated risk assessment to assist clinicians in decision-making.

Our work demonstrates that machine learning models can effectively analyze patient data to predict heart disease risk with high accuracy and interpretability. With the ongoing integration of AI in healthcare, such systems can play a vital role in **preventive diagnosis, remote patient monitoring, and clinical decision support**.

Future research will focus on **expanding datasets** with real-world hospital records, incorporating **additional biomarkers** such as ECG and genetic factors, and deploying **hybrid intelligent systems** that combine ensemble and deep learning for continuous, real-time cardiac health monitoring.

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